

Unit2-- BEVS

SECTION A — Two Mark Questions

Enzymes

Q1. What are enzymes? Write down any two applications of enzymes. [2]

Enzymes are biological macromolecules (mostly proteins, and some RNAs) that act as biocatalysts, accelerating biochemical reactions without being consumed in the process.

Two applications:

- **Medical:** Used in diagnostics, drug production, and enzyme replacement therapy (e.g., lactase for lactose intolerance).
- **Industrial:** Used in food processing (amylase in bread making) and textiles (cellulase for fabric softening).

Enzymes

Q2. What are the different ways of enzyme inhibition? [2]

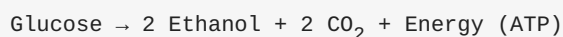
Enzyme inhibition can occur in three main reversible ways:

- **Competitive Inhibition:** The inhibitor competes with the substrate for the enzyme's active site.
- **Non-Competitive Inhibition:** The inhibitor binds to a different (allosteric) site, altering enzyme activity.
- **Uncompetitive Inhibition:** The inhibitor binds only to the enzyme-substrate complex.

Fermentation

Q3. Define fermentation. Write the equation for alcoholic fermentation. [2]

Fermentation is a metabolic process in which microorganisms (bacteria, yeast, or molds) convert sugars into simpler molecules to obtain energy, under anaerobic (oxygen-free) conditions.



Vitamin & Mineral Deficiency

Q4. Name the diseases that appear due to the deficiency of Vitamin A and Vitamin C. [2]

- **Vitamin A deficiency** causes **Night Blindness** (and, in severe cases, complete blindness).
- **Vitamin C deficiency** causes **Scurvy**, marked by swollen gums, joint pain, and skin problems.

Vitamin & Mineral Deficiency

Q5. Name the disease that appears due to the deficiency of Calcium and Iodine. [2]

- **Calcium deficiency** causes **Osteoporosis** in adults (brittle bones) and **Rickets** in children.
- **Iodine deficiency** causes **Goiter** (enlarged thyroid gland), due to impaired thyroid function.

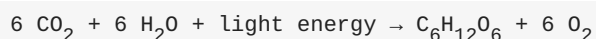
Bioenergy

Q6. Write down any two advantages of Bioenergy. [2]

- **Renewable:** Unlike fossil fuels, bioenergy is renewable as long as biomass is replenished through sustainable farming or forestry.
- **Carbon Neutral:** The CO_2 released on combustion is roughly equal to the CO_2 absorbed by the plants during growth.

Photosynthesis

Q7. Write the general equation of photosynthesis. Where does it mainly take place in a plant cell? [2]



Photosynthesis primarily takes place in the **chloroplasts** of plant cells, where the pigment **chlorophyll** absorbs sunlight.

Biofertilizers

Q8. What are biofertilizers? [2]

Biofertilizers are natural substances containing living microorganisms (bacteria, fungi, or algae) that enhance soil fertility and promote plant growth by increasing the availability of nutrients such as nitrogen, phosphorus, and potassium to plants. Example: *Rhizobium* (nitrogen-fixing bacteria).

Pesticides

Q9. What are pesticides? [2]

Pesticides are chemicals or biological agents used to control pests — including insects, weeds, fungi, and other organisms — that can damage crops, livestock, or structures.

Pesticides

Q10. Give two examples each of Insecticides, Herbicides, and Rodenticides. [2]

Type	Examples
Insecticides	Chlorpyrifos (organophosphate), Permethrin (pyrethroid)
Herbicides	Glyphosate (Roundup), 2,4-D
Rodenticides	Warfarin, Bromadiolone

Biological Pest Control

Q11. What do you mean by biological pest control? [2]

Biological pest control is an environmentally friendly and sustainable approach to managing pests using natural predators, parasites, or pathogens, instead of synthetic chemicals. It works by exploiting natural relationships such as predator-prey and host-parasite interactions. Example: Ladybugs (predators) controlling aphid populations.

Steroids

Q12. What are steroids? Name two examples. [2]

Steroids are a broad class of organic compounds with a common structure of four interconnected carbon rings. They regulate metabolism, inflammation, immune responses, and reproduction. Examples: **Cortisol** (a corticosteroid) and **Testosterone** (an anabolic steroid).

Vaccines

Q13. What are vaccines? Give two examples. [2]

Vaccines are biological preparations that stimulate the immune system to recognize and fight specific pathogens without causing the disease itself, typically by exposing the body to weakened or inactivated antigens. Examples: **COVID-19 vaccine** (mRNA type), **Polio vaccine** (inactivated virus type).

Vaccines / Human Health

Q14. Write the full form of HIV and AIDS. [2]

- **HIV** — Human Immunodeficiency Virus
- **AIDS** — Acquired Immunodeficiency Syndrome

Medicinal Plants

Q15. Mention the common uses of any two Indian medicinal plants. [2]

- **Tulsi (*Ocimum sanctum*):** Acts as an adaptogen, has antimicrobial and anti-inflammatory properties; used for cough, cold, and fever.
- **Neem (*Azadirachta indica*):** Antiseptic and antifungal; helps manage diabetes and is used in dental care.

Q16. Write down any two importances of medicinal plants in modern healthcare. [2]

- **Source of Pharmaceuticals:** Many modern drugs are derived from plants, e.g., aspirin (from willow bark) and quinine (from cinchona).
- **Economic Importance:** Medicinal plants contribute significantly to India's herbal industry and export market.

SECTION B — Seven Mark Questions

Enzymes

Q1. What are enzymes? Discuss their structure, characteristics, and significance as biocatalysts in biological processes. [7]

Definition: Enzymes are biological macromolecules — mostly proteins, though some are RNAs (ribozymes) — that act as biocatalysts, accelerating biochemical reactions without being consumed in the process.

Characteristics of Enzymes:

- **Specificity:** Highly specific for their substrates, explained by the lock-and-key or induced-fit models.
- **Catalytic Efficiency:** Accelerate reactions by lowering the activation energy.
- **Reusability:** Not consumed during the reaction and can be used repeatedly.
- **Optimal Conditions:** Function best under specific pH, temperature, and ionic strength.
- **Regulation:** Activity can be modulated by inhibitors, activators, or feedback mechanisms.

Mechanism of Action:

1. The substrate binds to the enzyme's active site, forming an enzyme-substrate complex.
2. The enzyme stabilizes the transition state, reducing the activation energy.
3. Products are released, and the enzyme is free to catalyze another reaction.

Types of Enzymes (based on reaction catalyzed): Oxidoreductases, Transferases, Hydrolases, Lyases, Isomerases, and Ligases.

Enzyme Inhibition: Competitive (inhibitor competes at the active site), Non-competitive (inhibitor binds an allosteric site), and Uncompetitive (inhibitor binds only the enzyme-substrate complex).

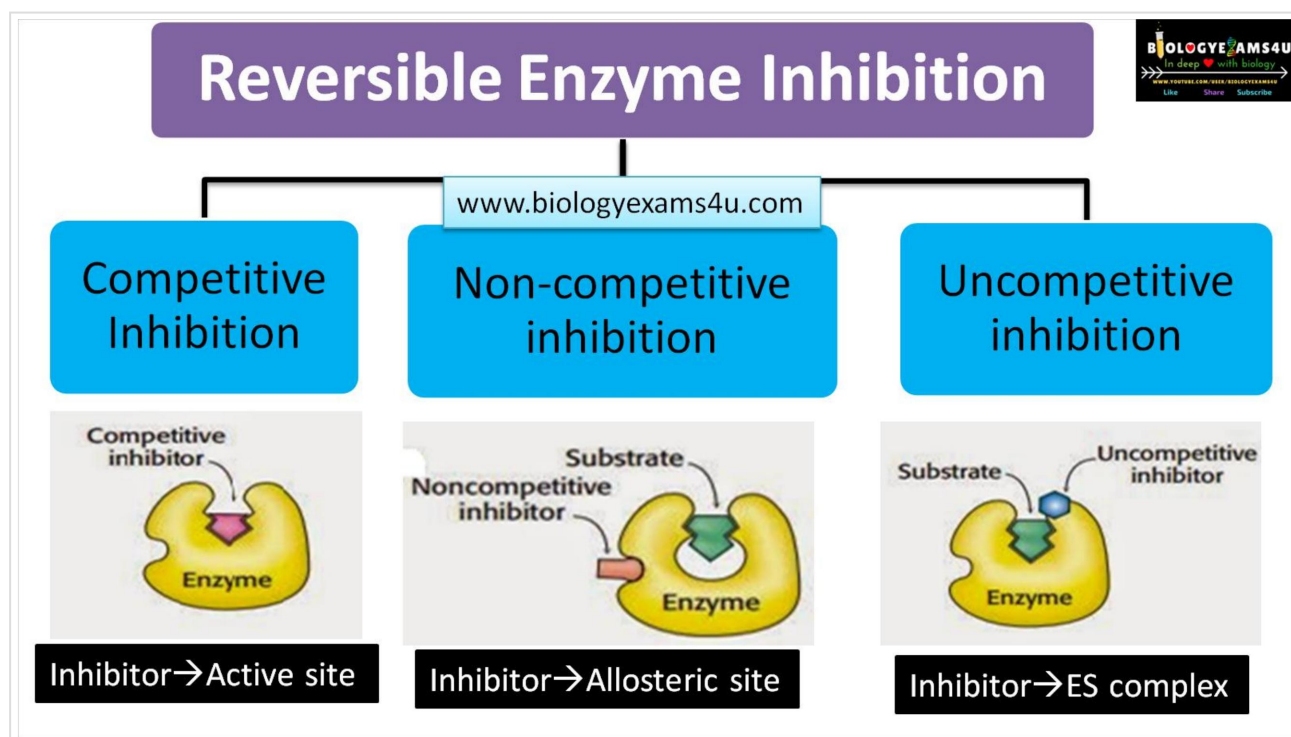


Fig: Reversible modes of enzyme inhibition — competitive, non-competitive, and uncompetitive.

Significance: Enzymes are essential for life processes such as metabolism, DNA replication, and energy production, and serve as efficient, eco-friendly alternatives to chemical catalysts in medicine, industry (food, textiles, biofuels), and research (e.g., DNA polymerase in PCR).

Fermentation

Q2. Define fermentation. Discuss the different steps of fermentation. [7]

Definition: Fermentation is a metabolic process in which microorganisms like bacteria, yeast, or molds convert sugars and other organic compounds into simpler molecules, often producing energy, under anaerobic (oxygen-free) conditions. It is widely used in food and beverage production and various industrial processes.

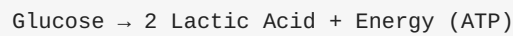
Steps of Fermentation:

1. **Sugar Breakdown (Glycolysis):** Glucose is split into two molecules of pyruvate, producing a small amount of ATP and NADH.
2. **Regeneration of NAD^+ :** Since glycolysis requires NAD^+ to continue, and oxygen is absent, the cell regenerates NAD^+ through fermentation.

3. Fermentation Pathways:

- **Alcoholic Fermentation (e.g., Yeast):** Pyruvate is converted into ethanol and CO_2 , regenerating NAD^+ . Used in brewing, wine-making, and baking. $\text{Glucose} \rightarrow 2 \text{ Ethanol} + 2 \text{ CO}_2 + \text{Energy (ATP)}$

- **Lactic Acid Fermentation (e.g., muscle cells, some bacteria):** Pyruvate is reduced to lactic acid, regenerating NAD^+ . Occurs in oxygen-deprived muscles (causing the "burn") and in dairy fermentation.



4. **Byproducts:** Ethanol and CO_2 (alcoholic fermentation) or lactic acid (lactic acid fermentation).

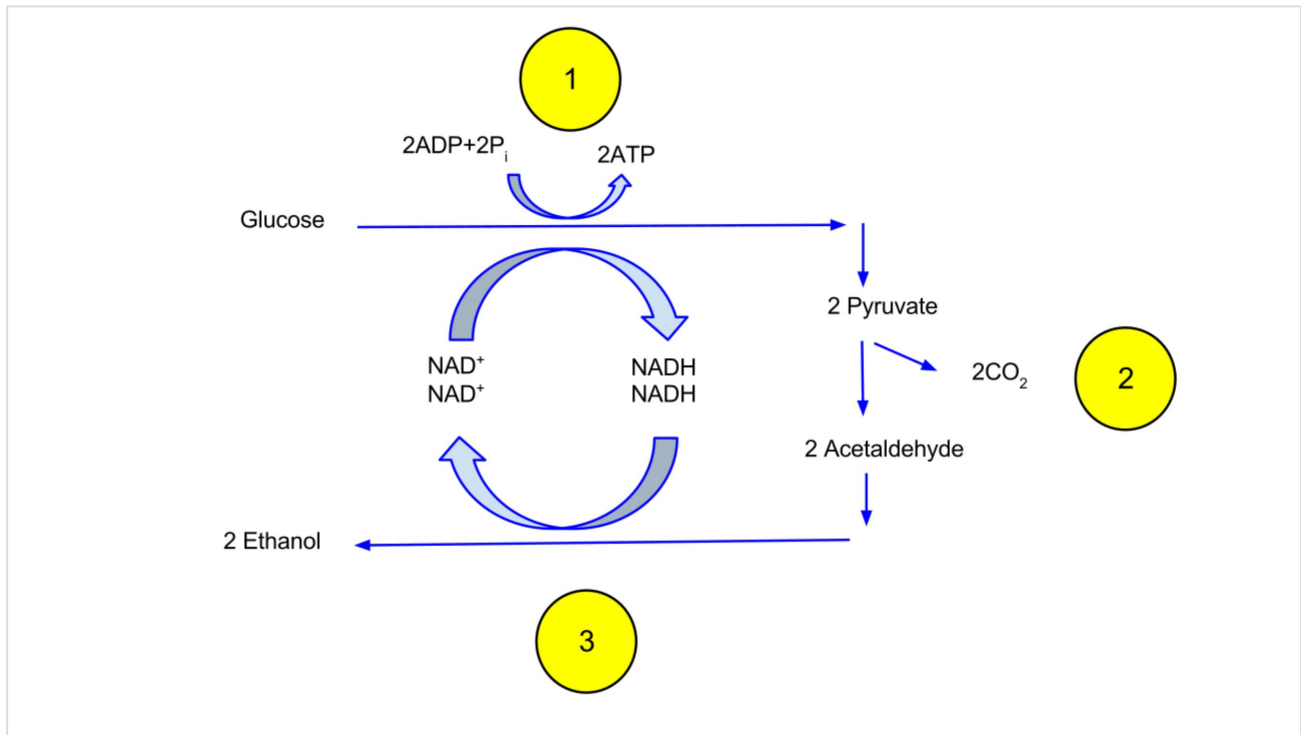


Fig: Pathway of alcoholic fermentation showing glycolysis, ATP/NADH cycling, and ethanol formation.

Applications: Bread making (CO_2 causes dough to rise), alcoholic beverages, dairy products (yogurt, cheese), and industrial production of biofuels, antibiotics, and other chemicals.

Q3. Define Bioenergy. What are the different forms of Bioenergy? Write down the different production processes and challenges of Bioenergy. [7]

Definition: Bioenergy is energy derived from biological sources such as plants, animals, and microorganisms (biomass). Unlike fossil fuels, it is renewable and can be replenished over time.

Forms of Bioenergy:

- **Biogas:** Produced by anaerobic decomposition of organic matter; mainly methane (CH_4), used for cooking, heating, or electricity.
- **Biofuels:** Liquid fuels for transportation — Ethanol (from fermenting sugars in crops) and Biodiesel (from plant oils or animal fats).
- **Biomass for Electricity and Heat:** Direct burning of biomass to generate power through steam turbines.
- **Biohydrogen:** Hydrogen gas produced from organic material, explored for fuel cells.

Production Processes:

- **Combustion:** Direct burning of biomass to release heat.
- **Anaerobic Digestion:** Breakdown of organic matter without oxygen to produce biogas.
- **Fermentation:** Conversion of crop sugars into ethanol by microorganisms.
- **Transesterification:** Converts vegetable oils/animal fats into biodiesel.
- **Gasification:** Biomass heated in low oxygen to produce syngas (H_2 , CO , CH_4).
- **Pyrolysis:** Biomass heated in absence of oxygen, yielding bio-oil, charcoal, and syngas.

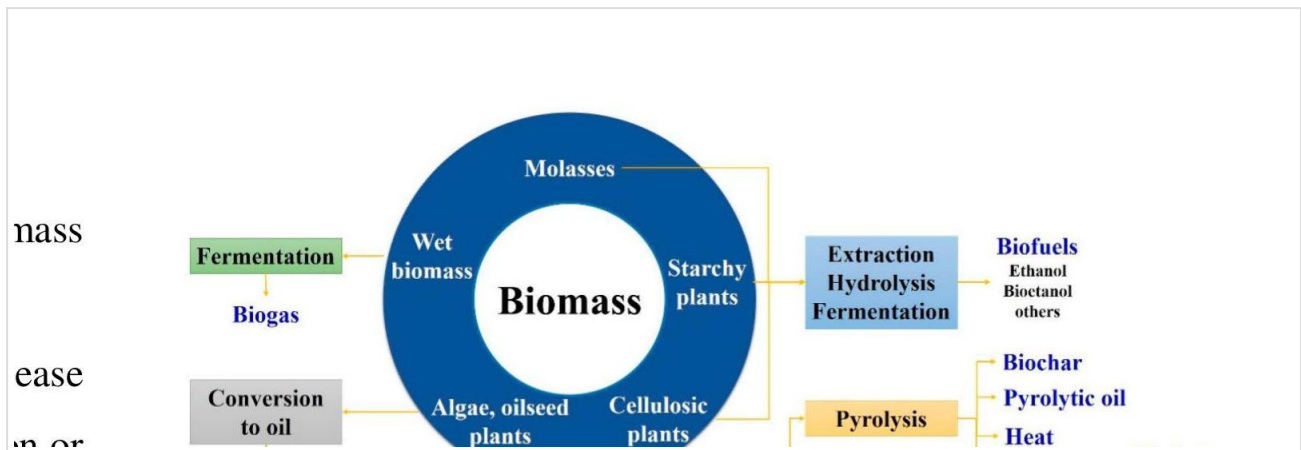


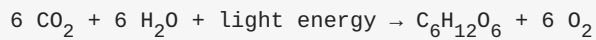
Fig: Biomass conversion routes into biofuels, biochar, pyrolytic oil, heat, and syngas.

Challenges of Bioenergy:

- **Land Use and Food Security:** Crops grown for biofuel may compete with food production.
- **Deforestation:** Unsustainable harvesting of forest biomass causes biodiversity loss.
- **Energy Efficiency:** Some methods (e.g., corn ethanol) have a low net energy gain.
- **Carbon Emissions:** The full lifecycle of production can still generate emissions.
- **Water Use:** Bioenergy crops often need large amounts of water.
- **Cost:** Production can be expensive and may require government subsidies.

Q4. What is photosynthesis? Discuss the different steps involved in photosynthesis. [7]

Definition: Photosynthesis is the process by which green plants, algae, and certain bacteria convert light energy into chemical energy, storing it in the form of glucose. It occurs mainly in the **chloroplasts**, where **chlorophyll** absorbs sunlight.



1. Light-Dependent Reactions (occur in the thylakoid membrane):

- 1. Light absorption:** Pigments (chlorophyll, carotenoids) absorb light.
- 2. Excitation of electrons:** Light energy excites electrons, transferring them to electron acceptors.
- 3. Transfer of electrons:** Electrons pass through a series of carriers, forming a proton gradient.
- 4. ATP synthesis:** The proton gradient drives ATP production through chemiosmosis; water is split, releasing O_2 .

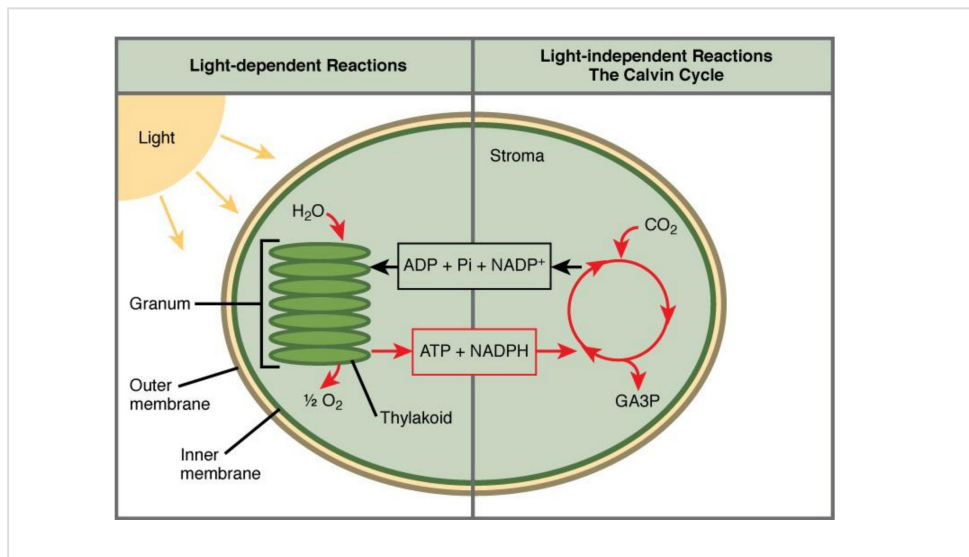


Fig: Chloroplast structure showing the light-dependent reactions (thylakoid) and the light-independent Calvin Cycle (stroma).

2. Light-Independent Reactions (Calvin Cycle) (occur in the stroma):

- 1. Carbon fixation:** CO_2 is fixed into a 3-carbon molecule (3-phosphoglycerate) by the enzyme RuBisCO.
- 2. Reduction:** 3-phosphoglycerate is reduced to glyceraldehyde-3-phosphate (G3P) using ATP and NADPH.
- 3. Regeneration of RuBP:** G3P regenerates ribulose-1,5-bisphosphate (RuBP) to continue the cycle.
- 4. Glucose production:** G3P molecules combine to form glucose.

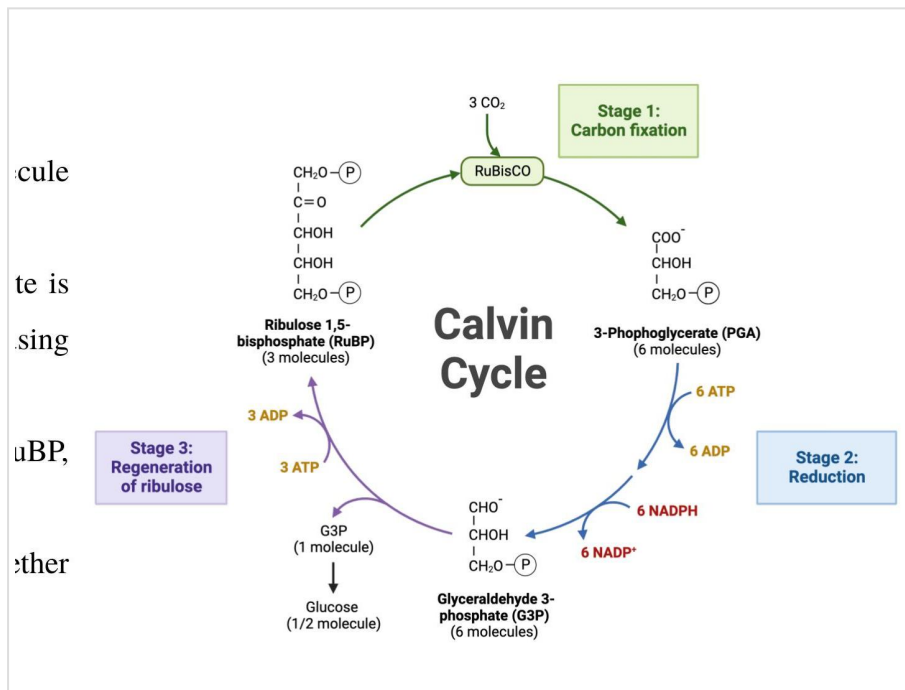


Fig: The Calvin Cycle — carbon fixation, reduction, and regeneration of ribulose.

Importance of Photosynthesis: Produces organic compounds that support life on Earth, releases oxygen into the atmosphere, and forms the basis of most food chains.

Biofertilizers

Q5. What are biofertilizers? How are biofertilizers applied? Write down the advantages and challenges of biofertilizers. [7]

Definition: Biofertilizers are natural substances containing living microorganisms (bacteria, fungi, or algae) that enhance soil fertility and promote plant growth by increasing nutrient availability.

Types of Biofertilizers:

- **Nitrogen-Fixing:** *Rhizobium* (symbiotic, in legumes), *Azotobacter* & *Azospirillum* (free-living).
- **Phosphorus-Solubilizing:** PSB (Phosphate Solubilizing Bacteria), Mycorrhizal Fungi (AMF).
- **Potassium-Solubilizing:** Certain bacteria and fungi that release bound potassium.
- **PGPR (Plant Growth-Promoting Rhizobacteria):** Promote growth via hormones or by suppressing soil pathogens.

Application Methods:

- **Seed Treatment:** Applied directly to seeds before planting.
- **Soil Application:** Mixed with soil to boost microbial activity.
- **Root Inoculation:** Applied directly to plant roots.
- **Foliar Spray:** Sprayed on leaves to enhance growth and pathogen resistance.

Advantages:

- **Eco-friendly:** Reduce dependence on chemical fertilizers.
- **Sustainable Farming:** Improve soil health and biodiversity.
- **Cost-effective:** Lower long-term farming input costs.
- **Soil Health:** Enhance soil structure and microbial diversity.

Challenges:

- **Storage & Shelf Life:** Require proper storage; limited shelf life due to living organisms.
- **Environmental Conditions:** Efficacy depends on soil temperature, pH, and moisture.

Q6. What are pesticides? Write down four advantages and four disadvantages of pesticides. [7]

Definition: Pesticides are chemicals or biological agents used to control pests — insects, weeds, fungi, and other organisms — that damage crops, livestock, or structures.

Types of Pesticides:

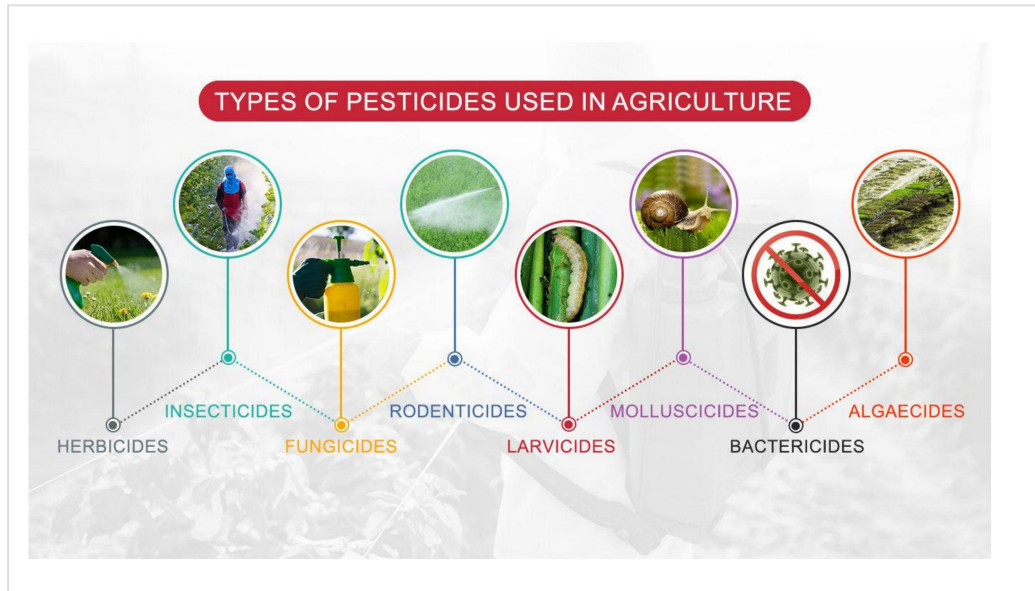


Fig: Major types of pesticides used in agriculture.

- **Insecticides** (e.g., chlorpyrifos, permethrin) — control insect pests.
- **Herbicides** (e.g., glyphosate, 2,4-D) — control unwanted weeds.
- **Fungicides** (e.g., copper sulfate) — control fungal diseases.
- **Rodenticides** (e.g., warfarin) — control rodent populations.
- **Bactericides** and **Nematicides** — control harmful bacteria and nematodes respectively.

Four Advantages:

1. **Increased Crop Yield:** Controls pests, ensuring healthier crops and reduced damage.
2. **Protection Against Disease:** Protects plants and animals from disease-causing organisms.
3. **Economic Benefits:** Higher yields and reduced crop loss improve farmers' profits.
4. **Faster Action:** Acts quickly to control pest outbreaks.

Four Disadvantages:

1. **Environmental Pollution:** Runoff can contaminate soil, water, and air.
2. **Non-target Effects:** Can harm beneficial organisms like bees, birds, and aquatic life.
3. **Human Health Risks:** Toxicity from exposure, and long-term effects like cancer or reproductive issues.
4. **Resistance & Biodiversity Loss:** Overuse leads to pest resistance and reduced biodiversity.

Q7. What are Steroids? Discuss the function of different kinds of Steroids. Write down the different biological functions of Steroids. [7]

Definition: Steroids are a broad class of organic compounds with a common structure of four interconnected carbon rings. They perform various functions in the body, from regulating metabolism to acting as signaling molecules.

Types of Steroids and Their Functions:

- **Corticosteroids:** Produced by the adrenal glands; help manage stress and regulate inflammation, metabolism, and immune responses.
 - **Cortisol:** Regulates metabolism, reduces inflammation, helps the body respond to stress.
 - **Aldosterone:** Regulates sodium and potassium levels, maintaining blood pressure.
- **Anabolic Steroids:** Synthetic derivatives of testosterone; promote muscle growth, improve athletic performance, and play a role in male sexual development.
 - **Testosterone:** Primary male sex hormone, crucial for muscle growth and overall male health.
 - **Synthetic derivatives** (e.g., nandrolone, stanozolol): Mimic testosterone to increase muscle mass and strength.

Biological Functions of Steroids:

- **Metabolic Regulation:** Corticosteroids regulate glucose production and fat storage.
- **Immune Function:** Corticosteroids modulate the immune system and control inflammation.
- **Water and Salt Balance:** Aldosterone controls fluid balance, blood pressure, and electrolytes.
- **Development and Reproduction:** Sex hormones (estrogen, progesterone, testosterone) regulate sexual characteristics, fertility, and reproduction.

Uses: Corticosteroids treat inflammatory/autoimmune conditions (arthritis, asthma, lupus); anabolic steroids treat muscle-wasting diseases and hormone deficiencies; sex hormones are used in hormone replacement therapy and contraceptives. (Note: misuse of anabolic steroids for performance enhancement is illegal in most sports and carries serious health risks.)

Q8. What is biological pest control? Discuss its types with examples, and mention its applications. [7]

Definition: Biological pest control is an environmentally friendly, sustainable approach to managing pests using natural predators, parasites, or pathogens, instead of synthetic chemicals like pesticides. It works by exploiting natural relationships between organisms.

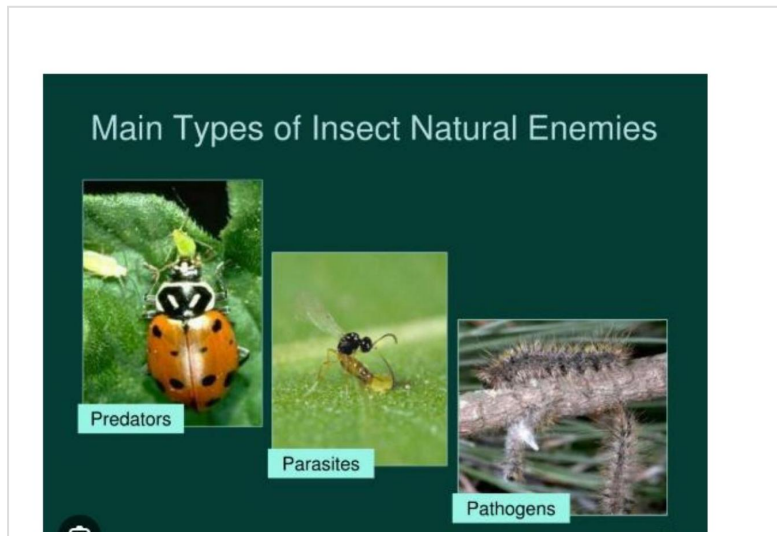


Fig: Main types of insect natural enemies — predators, parasites, and pathogens.

Types of Biological Pest Control:

- **Predators:** Organisms that hunt and kill pests for food.
Examples: **Ladybugs** (prey on aphids), **Lacewing larvae** (consume aphids, mealybugs), **Praying mantises** (eat a wide variety of insects).
- **Parasites:** Live in or on a host organism, ultimately killing it to complete their life cycle.
Examples: **Parasitic (Trichogramma) wasps** (lay eggs inside pest insects), **Nematodes** (parasitize soil-dwelling pests like grubs).
- **Pathogens:** Microorganisms (bacteria, fungi) that infect and kill pests.
Examples: **Bacillus thuringiensis (Bt)** — toxic to caterpillars; **Beauveria bassiana** — a fungus that infects termites, whiteflies, and beetles.

Applications of Biological Control:

- **Agriculture:** Controls pests on vegetables, fruits, and field crops.
- **Horticulture & Gardening:** Home gardeners use predators like ladybugs.
- **Greenhouses:** Used in enclosed spaces to control whiteflies and aphids.
- **Forestry:** Manages pests threatening forests, such as the pine beetle.

Q9. Define Vaccines. How do Vaccines work? Write down the importance of vaccines. [7]

Definition: Vaccines are biological preparations designed to provide immunity to specific diseases. They stimulate the immune system to recognize and fight pathogens (bacteria, viruses, or parasites) without causing the disease itself.

How Vaccines Work:

- **Immune System Activation:** Vaccines contain weakened or inactivated pathogens (or parts of them) that trigger an immune response without causing illness.
- **Antigen Exposure:** Antigens in the vaccine cause the immune system to produce antibodies that can neutralize the pathogen.
- **Memory Cells:** The immune system produces memory B-cells and T-cells that "remember" the pathogen, enabling a faster, stronger response on future exposure.

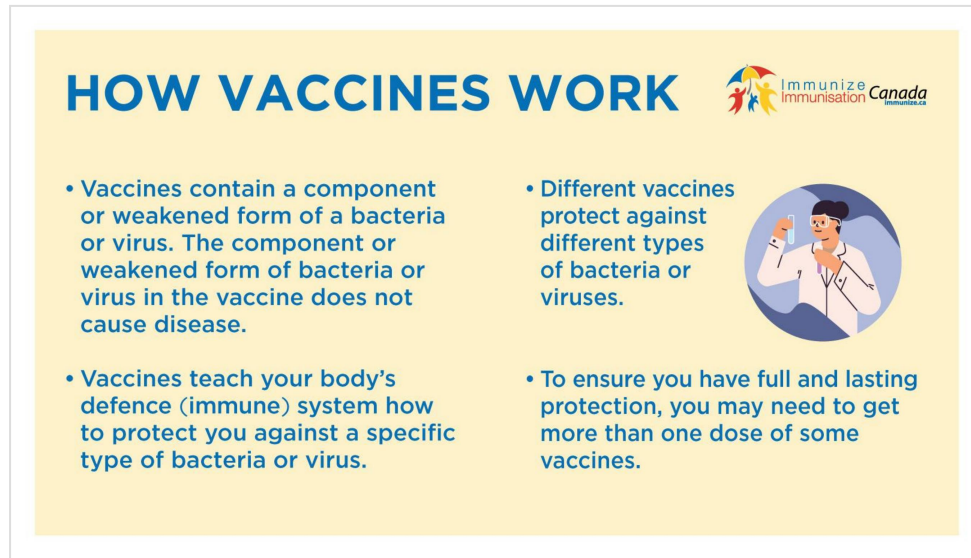


Fig: How vaccines work — training the immune system without causing disease.

Types of Vaccines: RNA, DNA, Viral vector, Virus-like particle, Protein sub-unit, and Inactivated virus vaccines.

Importance of Vaccines:

- **Prevention of Disease:** Prevent the spread of infectious diseases, reducing illness in the population.
- **Herd Immunity:** Protects those who cannot be vaccinated (infants, immunocompromised, elderly) when a large portion of the population is vaccinated.
- **Eradication of Diseases:** Led to the eradication of smallpox and near-elimination of polio.
- **Reduced Healthcare Costs:** Prevents hospitalizations and long-term care needs.

Challenges in Vaccination: Vaccine hesitancy due to misinformation, limited access in underdeveloped regions, and difficulty developing vaccines for rapidly mutating pathogens (e.g., HIV, influenza).

Q10. Mention the common uses of different Indian medicinal plants, and write down the importance of medicinal plants in modern healthcare. [7]

India is renowned for its rich biodiversity and traditional knowledge of medicinal plants, integral to **Ayurveda, Siddha, and Unani** systems of medicine.

Plant	Key Uses
Tulsi (<i>Ocimum sanctum</i>)	Adaptogen; antimicrobial and anti-inflammatory; treats cough, cold, bronchitis, fever; boosts immunity.
Neem (<i>Azadirachta indica</i>)	Antiseptic/antifungal for skin diseases; lowers blood sugar; used in dental care and as insect repellent.
Amla (<i>Phyllanthus emblica</i>)	Boosts immunity and digestion; antioxidant; promotes hair growth; supports heart health.
Ashwagandha (<i>Withania somnifera</i>)	Reduces stress/anxiety; enhances strength and immunity; supports brain function; treats arthritis.
Turmeric (<i>Curcuma longa</i>)	Anti-inflammatory and antimicrobial; wound healing; manages arthritis; supports liver health.
Ginger (<i>Zingiber officinale</i>)	Aids digestion; relieves nausea; anti-inflammatory, helps manage arthritis.
Aloe Vera (<i>Aloe barbadensis</i>)	Soothes skin irritation and wound healing; aids digestion; natural laxative.
Brahmi (<i>Bacopa monnieri</i>)	Enhances memory and cognition; reduces anxiety; manages epilepsy.
Arjuna (<i>Terminalia arjuna</i>)	Strengthens heart muscles; manages blood pressure and cholesterol; wound healing.
Shatavari (<i>Asparagus racemosus</i>)	Promotes reproductive health in women; galactagogue (enhances milk production); boosts immunity.

Importance of Medicinal Plants in Modern Healthcare:

- 1. Source of Pharmaceuticals:** Many modern drugs are derived from plants, e.g., aspirin (from willow bark) and quinine (from cinchona).
- 2. Holistic Healing:** Used in alternative medicine systems like Ayurveda for treating diseases with fewer side effects.
- 3. Economic Importance:** Contribute significantly to India's herbal industry and export market.

The Ministry of Ayush has established a **National Medicinal Plant Board** to promote cultivation, conservation, and sustainable management of medicinal plants through schemes such as the Central Sector Scheme for Conservation, Development and Sustainable Management of Medicinal Plants.